

AI-Based MOVIE BOOKING SYSTEM

Requirements Document

1. Project Objective

Build a web-based Movie Booking System with AI that enables customers to discover movies, select theatres and shows, reserve seats, make payments, receive e-tickets, and obtain personalized AI-powered recommendations and assistance. Administrators and theatre managers can manage movies, theatres, screens, shows, pricing, bookings, offers, and analytics.

2. User Roles

Role	Responsibilities
Customer	Browse movies, book tickets, make payments, cancel tickets, view history, receive recommendations.
Theatre Manager	Manage screens, seats, shows, pricing, occupancy, and theatre-specific information.
Admin	Manage users, movies, theatres, shows, offers, reports, and system configuration.
AI Recommendation Engine	Analyze user preferences and generate personalized movie recommendations.
AI Chatbot	Answer movie, show, booking, theatre, and availability-related questions.

3. Customer Requirements

3.1 Registration and Login

- Register using name, email, mobile number, and password.
- Login and logout securely.
- Reset and change password.
- Update customer profile.
- Manage preferred language/location.
- View booking history.

3.2 Movie Search

- Search by movie name, genre, language, actor, director, release date, rating, theatre, or location.
- Filter movies by availability, language, genre, rating, and date.
- Support natural-language search such as: "Action movies in English available tonight."

4. Movie Management

- Add, update, deactivate, and view movies.
- Maintain movie name, description, genre, language, duration, certification, release date, cast, director, rating, poster, and trailer URL.
- Movie status: Upcoming, Now Showing, Coming Soon, Completed, or Inactive.

5. Theatre and Screen Management

- Create and manage theatres, addresses, locations, facilities, and contact details.
- Create multiple screens under a theatre.
- Configure screen type such as standard or IMAX.
- Define seat rows, seat numbers, categories, and prices.
- Seat categories may include Regular, Premium, Recliner, and VIP.

6. Show Management

- Create, update, and cancel movie shows.
- Assign movie, theatre, screen, date, start time, end time, and ticket pricing.
- Prevent overlapping shows on the same screen.
- Allow theatre managers to view occupancy and show performance.

7. Seat Booking

The application should provide a visual seat layout with Available, Selected, Booked, and Blocked states.

- Customers can select one or more available seats.
- Selected seats should be temporarily locked during payment.
- Seat locks should expire automatically after a configurable period, such as five minutes.
- The system must prevent double booking of the same seat for the same show.
- Failed or expired payments must release locked seats.

8. Booking Management

- Create a booking containing booking ID, customer, movie, theatre, screen, show, seats, quantity, price, discount, tax, and final amount.
- Maintain booking timestamp and booking status.
- Booking statuses: PENDING, CONFIRMED, CANCELLED, EXPIRED, and COMPLETED.
- Allow customers to view current and previous bookings.

9. Payment Management

- Integrate a mock payment gateway for the academic project.
- Maintain payment ID, booking ID, amount, payment method, transaction reference, date, and status.
- Payment statuses: INITIATED, SUCCESS, FAILED, and REFUNDED.
- Booking confirmation should occur only after successful payment.

10. E-Ticket and QR Code

- Generate an e-ticket after successful payment.
- Ticket should contain booking ID, movie, theatre, screen, date, time, seats, amount, and QR code.
- QR code can be used for simulated theatre-entry validation.

11. Cancellation and Refund

- Allow cancellation according to configurable cancellation rules.

- Support full, partial, or no refund based on time before the show.
- Maintain cancellation and refund transaction details.
- Cancelled bookings must not remain available for normal ticket usage.

12. AI Features

12.1 Personalized Movie Recommendation

- Analyze booking history, genres, languages, ratings, actors, directors, preferred show timings, and search history.
- Recommend movies based on customer interests.
- Exclude unavailable, inactive, or past movies from recommendations.

12.2 Hybrid Recommendation Engine

- Content-based filtering: recommend movies similar to previously viewed/booked movies.
- Collaborative filtering: recommend movies based on users with similar preferences.
- Combine both approaches using a hybrid recommendation score.
- Store recommendation results for analytics and evaluation.

12.3 AI Movie and Booking Chatbot

- Answer questions about movies, shows, theatres, seats, prices, and bookings.
- Support intents such as SEARCH_MOVIE, SEARCH_SHOW, FIND_THEATRE, CHECK_SEATS, GET_BOOKING, CANCEL_BOOKING, and MOVIE_RECOMMENDATION.
- Use backend APIs to retrieve real-time application data.
- Apply authentication and authorization when accessing customer-specific information.

12.4 AI Demand Prediction

- Predict expected ticket demand for a movie/show.
- Use historical bookings, movie popularity, genre, day of week, show time, holidays, theatre, and occupancy.
- Display predicted demand as Low, Medium, or High.
- Display predicted occupancy percentage for theatre managers.

12.5 Dynamic Pricing Simulation

- Calculate simulated ticket pricing based on predicted demand.
- Low demand: normal price.
- Medium demand: configurable increase.
- High demand: configurable increase.
- Use simulation only for academic purposes unless a real pricing policy is explicitly required.

12.6 Personalized Offers

- Identify customer preferences and booking patterns.
- Generate personalized offers based on customer segment and behaviour.
- Track offer eligibility, validity, usage, and conversion.

12.7 Review Sentiment Analysis

- Analyze customer reviews using an NLP/AI model.
- Classify reviews as Positive, Neutral, or Negative.
- Provide aggregate sentiment statistics to administrators.

13. Notification Management

- Booking confirmation.
- Payment success/failure.
- Cancellation and refund notifications.
- Upcoming-show reminders.
- Movie recommendations.
- Personalized promotional offers.

14. Admin Dashboard

- Display total users, movies, theatres, shows, bookings, revenue, cancellations, and occupancy rate.
- Provide daily/weekly/monthly booking and revenue charts.
- Show movie-wise bookings and theatre-wise revenue.
- Display popular genres and AI recommendation conversion.
- Provide audit and operational reports.

15. Theatre Manager Dashboard

- Today's and upcoming shows.
- Seat occupancy and revenue.
- Cancelled tickets.
- Popular movies.
- Demand prediction.
- Simulated dynamic pricing recommendations.
- Screen utilization.

16. Suggested Database Tables

Table	Purpose
users	User accounts
roles	Application roles
user_roles	User-role mapping
movies	Movie information
genres	Genre master
movie_genres	Movie-genre mapping
actors	Actor information
movie_cast	Movie-cast mapping
directors	Director information
theatres	Theatre information
screens	Theatre screens
seats	Screen seats
shows	Movie show schedules
show_seats	Show-specific seat availability
bookings	Booking header
booking_seats	Booked seats
payments	Payment information
tickets	E-ticket information
reviews	Customer reviews
offers	Offers
user_offers	Personalized offers

notifications	Notifications
recommendations	AI recommendations
user_preferences	Customer preferences
search_history	Search activity
cancellation	Cancellation/refund
audit_logs	System audit information

17. Suggested REST APIs

Authentication

- POST /api/auth/register
- POST /api/auth/login
- POST /api/auth/forgot-password

Movies

- GET /api/movies
- GET /api/movies/{id}
- POST /api/movies
- PUT /api/movies/{id}
- DELETE /api/movies/{id}

Theatres

- GET /api/theatres
- POST /api/theatres
- PUT /api/theatres/{id}

Shows

- GET /api/shows
- GET /api/shows/{id}
- POST /api/shows
- PUT /api/shows/{id}
- DELETE /api/shows/{id}

Seats

- GET /api/shows/{showId}/seats
- POST /api/shows/{showId}/seats/lock
- POST /api/shows/{showId}/seats/release

Bookings

- POST /api/bookings
- GET /api/bookings/{id}
- GET /api/users/{userId}/bookings
- POST /api/bookings/{id}/cancel

Payments

- POST /api/payments

- GET /api/payments/{id}

AI

- GET /api/ai/recommendations/{userId}
- POST /api/ai/chat
- GET /api/ai/demand/{showId}
- GET /api/ai/offers/{userId}
- POST /api/ai/review/sentiment

18. Recommended Technology Stack

- Backend: Java 17/21, Spring Boot, Spring MVC, Spring Data JPA/Hibernate, Spring Security, JWT, Bean Validation, Maven.
- Database: MySQL or PostgreSQL.
- Frontend: Angular. For an advanced capstone, Angular/React with Spring Boot REST APIs is recommended.
- AI: Spring AI and/or a Python ML service using a recommendation or NLP model; an LLM API can be used for chatbot capabilities.
- Engineering tools: Git/GitHub, Swagger/Open API.
- Testing: JUnit, Mockito, integration/API testing, and frontend unit testing.

19. Non-Functional Requirements

- Security: password hashing, JWT authentication, role-based authorization, input validation, SQL injection protection, CORS configuration, secure error handling, and audit logging.
- Performance: pagination, database indexing, caching where appropriate, optimized queries, and efficient data loading.
- Reliability: transaction management, seat-lock timeout, payment failure handling, booking rollback, and global exception handling.
- Maintainability: layered architecture, DTOs, centralized exception handling, logging, API documentation, and clean coding standards.

20. Important Business Rules

1. A seat cannot be booked twice for the same show.
2. A customer cannot book a past show.
3. Booking requires successful payment.
4. Failed payment must release locked seats.
5. Seat locks expire automatically.
6. Cancelled bookings cannot be used for theatre entry.
7. Refund amount depends on the cancellation policy.
8. A show cannot overlap another show on the same screen.
9. A screen cannot be assigned to overlapping shows.
10. Only authorized users can manage movies, theatres, shows, and pricing.

11. AI recommendations should not recommend unavailable or past movies.
12. AI chatbot must respect logged-in user permissions.
13. Administrative actions must be recorded in audit logs.

21. Recommended Advanced Capstone Scope

For an advanced capstone implementation, the recommended scope is to implement five core AI capabilities: personalized movie recommendation, AI movie/booking chatbot, demand prediction, personalized offers, and review sentiment analysis. These should be integrated with the Spring Boot transactional booking system rather than treated as isolated AI demonstrations.